

Ch9 Support Vector Machines (only for classif.)

pg 1

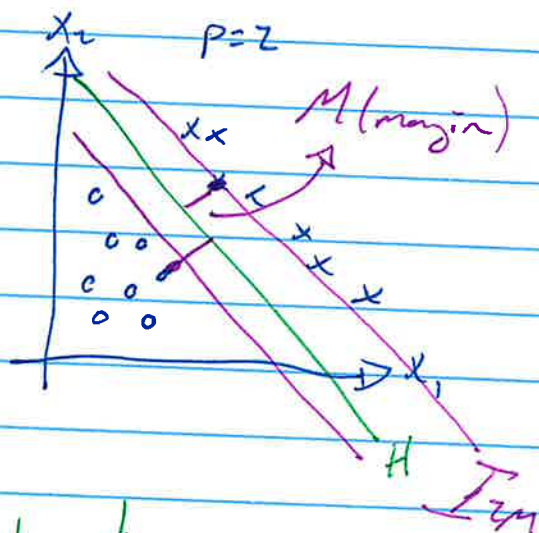
SVM came from stats (60s) & grew in CS (90s) as 1st ML classif. algs.

SVM comparable to logistic regression.

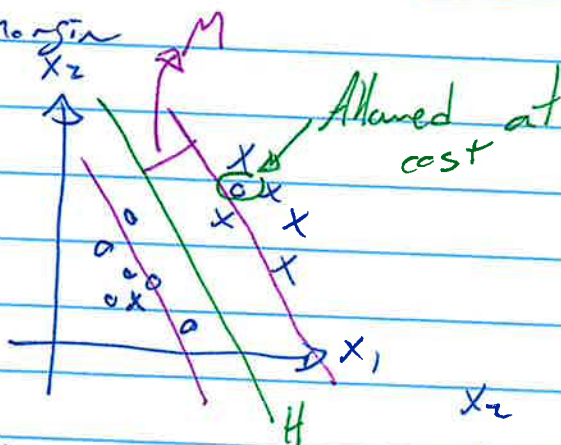
Build to ideas of SVMs via 3 steps:

V1 ^{Best} Max margin classifier
↳ hyper-plane separator

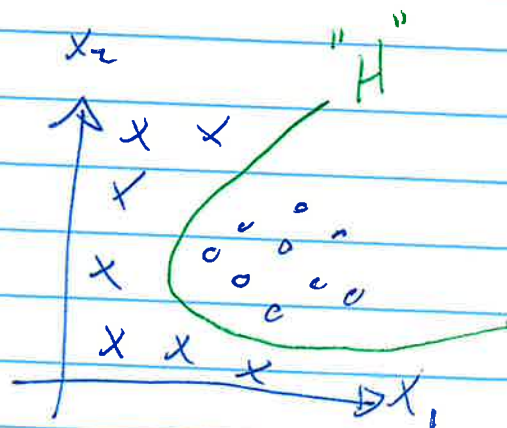
Review.



V2 Support vector classifier
↳ Soft margin



V3 Support vector machine
↳ Kernel trick



Allow for train misclass.

Allow for non-linear decs boundary

$x_1, x_2, \dots, x_p$
 What is a hyper-plane? of $\mathbb{R}^p$ (p-1) dimension "subspace" $\vec{v} \notin H$

2D Line in x_1 - x_2 plane

General form: $\vec{\beta}_0 + \vec{\beta}_1 x_1 + \vec{\beta}_2 x_2 = 0$

Same as L.R.
 $\vec{\beta} \cdot (\vec{x} - \vec{x}_0) = 0$

$$\vec{\beta} \cdot \vec{x} - \vec{\beta} \cdot \vec{x}_0 = 0$$

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 0$$

$$3x_1 + 2x_2 - 6 = 0$$

$$3x + 2y - 6 = 0$$

$$y = -\frac{3}{2}x + 3$$

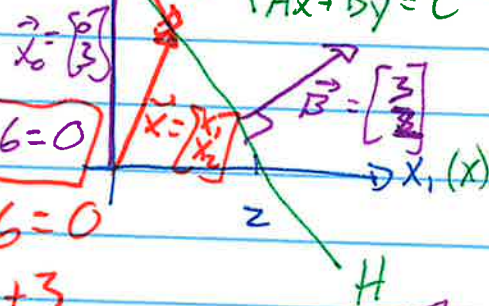
(y) $m = -\frac{3}{2}$ pg 2

$$y = -\frac{3}{2}x + 3$$

$$y = mx + b$$

$$y - y_1 = m(x - x_1)$$

$$Ax + By = C$$



3D Planes in 3-space

$\vec{\beta}$ orth to H

$$\vec{\beta}_0 + \vec{\beta}_1 x_1 + \vec{\beta}_2 x_2 + \vec{\beta}_3 x_3 = 0$$

How to get formula for H ?

Use vectors! $\mathbb{R}^p$ same.

Need 2 things:

① Normal vector: $\vec{\beta}$

② Point on H : $\vec{x}_0$

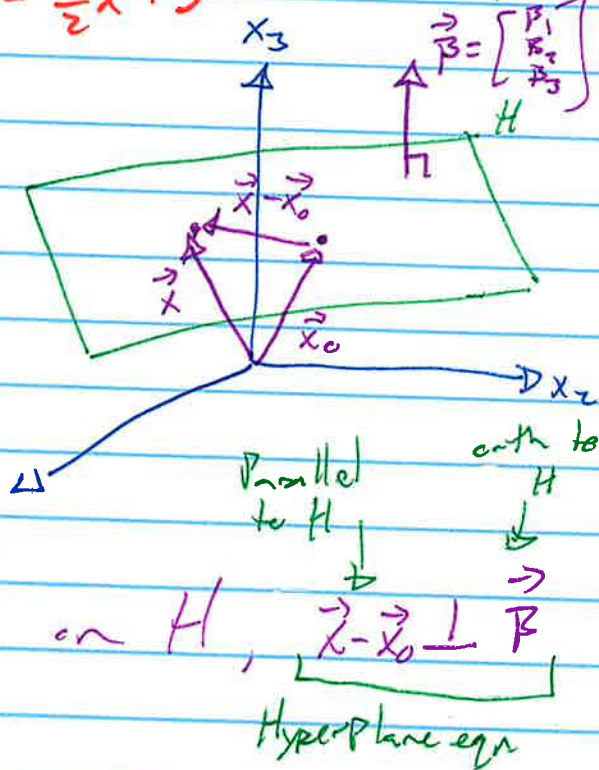
Then for any $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ on H , $\vec{x} - \vec{x}_0 \perp \vec{\beta}$

$$\vec{\beta} \cdot (\vec{x} - \vec{x}_0) = 0$$

$$\vec{\beta} \cdot \vec{x} - \vec{\beta} \cdot \vec{x}_0 = 0$$

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} - \vec{\beta} \cdot \vec{x}_0 = 0$$

$$\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_0 = 0$$



Hyperplane in p -dims ($\mathbb{R}^p$):

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p = 0$$

⊛ $\beta_0 + \vec{\beta} \cdot \vec{x} = 0$, $\vec{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$

↳ Also write as: $\beta_0 + \vec{\beta}^T \vec{x} = 0$

Define: $f(\vec{x}) = \beta_0 + \vec{\beta} \cdot \vec{x}$

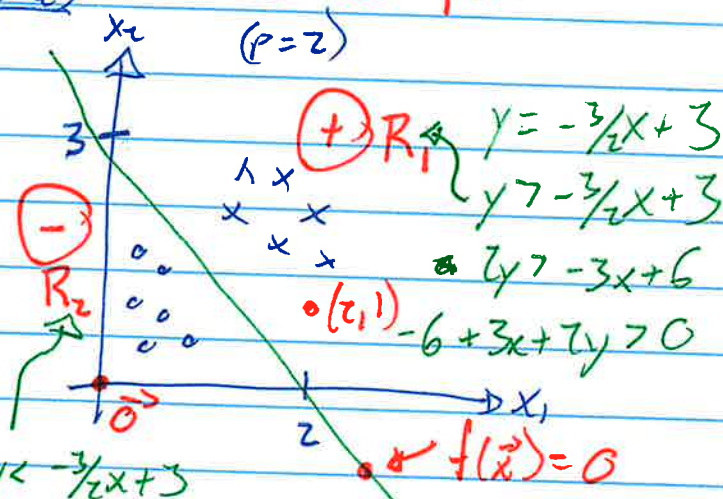
$\beta_0 + \langle \vec{\beta}, \vec{x} \rangle = 0$

2 key features of hyperplanes General dot prod

① H separates $\mathbb{R}^p$ + check which side on

$p=2$

- $f(\vec{x}) = 0$ on H
- $f(\vec{x}) > 0$ in $R_1 \oplus$
- $f(\vec{x}) < 0$ in $R_2 \ominus$



↳ Classification rule

"Prob" of classif.

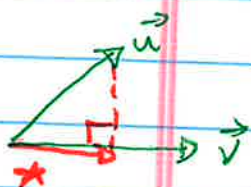
② Formula for dist to H .

Via vector proj.

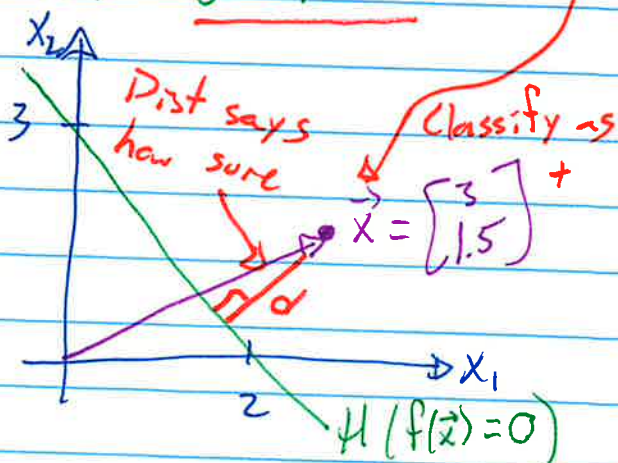
$$d = \frac{f(\vec{x})}{\|\vec{\beta}\|} = \frac{\beta_0 + \vec{\beta} \cdot \vec{x}}{\|\vec{\beta}\|}$$

Typical $\vec{\beta}$ as $\|\vec{\beta}\| = 1$

* $\text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}$

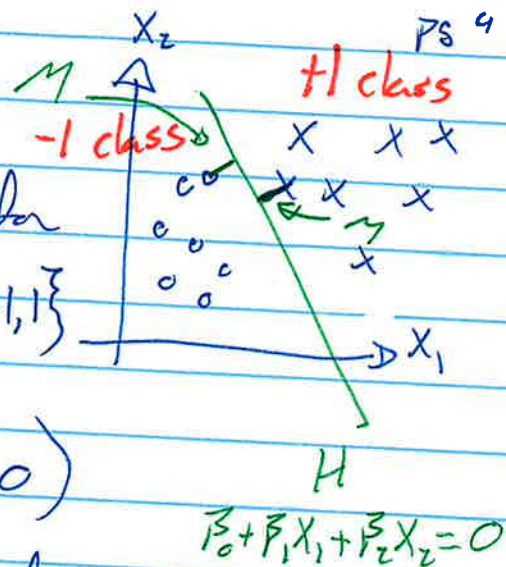


$$f(\vec{x}) = -6 + 3x_1 + 2x_2 = 0$$



Return to classification:

If can separate my train data
 $(\vec{X}_i, y_i)$, $\vec{X}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix}$, $y_i \in \{-1, 1\}$
 $i=1, \dots, n$



by hyperplane $H(\vec{\beta}_0 + \vec{\beta} \cdot \vec{X} = 0)$

Find $H(\vec{\beta}_0 + \vec{\beta})$ such that

many
H

$$\begin{cases} \textcircled{1} \vec{\beta}_0 + \vec{\beta} \cdot \vec{X}_i > 0 & \text{for } y_i = 1 \\ \textcircled{2} \vec{\beta}_0 + \vec{\beta} \cdot \vec{X}_i < 0 & \text{for } y_i = -1 \end{cases} \quad \text{all } i$$

Write compactly as

Same as $\textcircled{1}$

$$\rightarrow y_i (\vec{\beta}_0 + \vec{\beta} \cdot \vec{X}_i) > 0 \quad \text{all } i$$

$y_i = 1$ $\textcircled{1}$ Max margin classifier

$y_i = -1$ $\textcircled{2}$ Choose H such $\textcircled{1}$ true (separates classes)
 + maximizes margin M .

MMC
 optimization
 prob

$$\begin{cases} \max_{\vec{\beta}_0, \vec{\beta}} M \\ \text{Subject to} \\ y_i (\vec{\beta}_0 + \vec{\beta} \cdot \vec{X}_i) > M \end{cases}$$

to relax
 cannot misclassify
 hyperplane separ

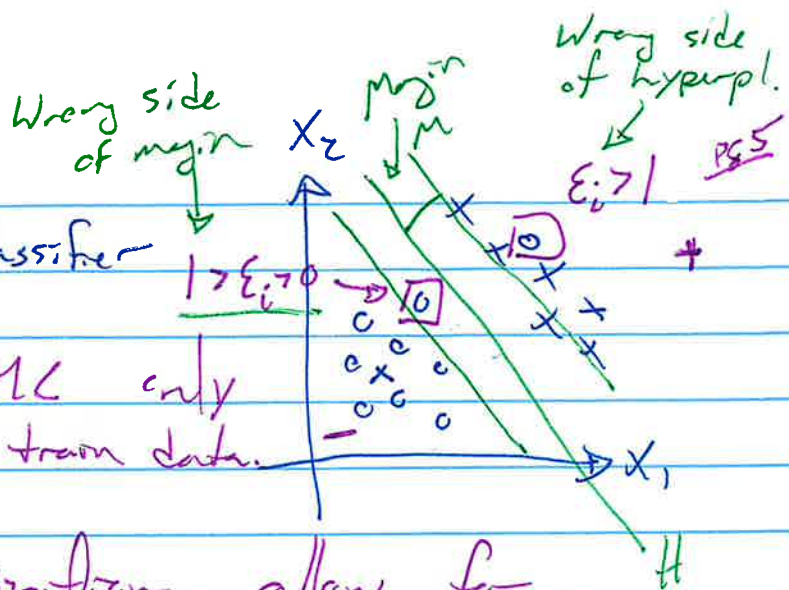
$$M_i = \frac{f(\vec{X}_i)}{\|\vec{\beta}\|} = \frac{\vec{\beta}_0 + \vec{\beta} \cdot \vec{X}_i}{\|\vec{\beta}\|}$$

$$M = \min_{1 \leq i \leq n} M_i$$

Simplify
 try choose
 $\|\vec{\beta}\| = 1$

V2 Support vector classifier

Issue to resolve: MMC only
works on separable train data.



Adjust: Semish optimization, allow for
misclassification

$$\text{SVC optimize } \left\{ \begin{array}{l} \max_{\vec{\beta}_0, \vec{\beta}, M_i} \mathcal{M} \\ \text{subject to} \end{array} \right. \quad \left(\mathcal{M}_i = f(\vec{x}) = \vec{\beta}_0 + \vec{\beta} \cdot \vec{x} \right)$$

$$\textcircled{1} \quad \|\vec{\beta}\| = 1$$

$$\textcircled{2} \quad y_i (\vec{\beta}_0 + \vec{\beta} \cdot \vec{x}) \geq \mathcal{M}_i (1 + \underbrace{\epsilon_i}_{\text{Relax}})$$

$$\textcircled{3} \quad \epsilon_i \geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C \quad \text{Relaxed margin}$$

Cost C is a tuning
param found via CV.

Cost of
relaxing.

Ch 9 Support vector machines (classif.)

$$\frac{f(\vec{x})}{\|\vec{b}\|} = \text{signed distance to } H \quad 1/6$$

Recap: $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$, $y \in \{-1, 1\}$

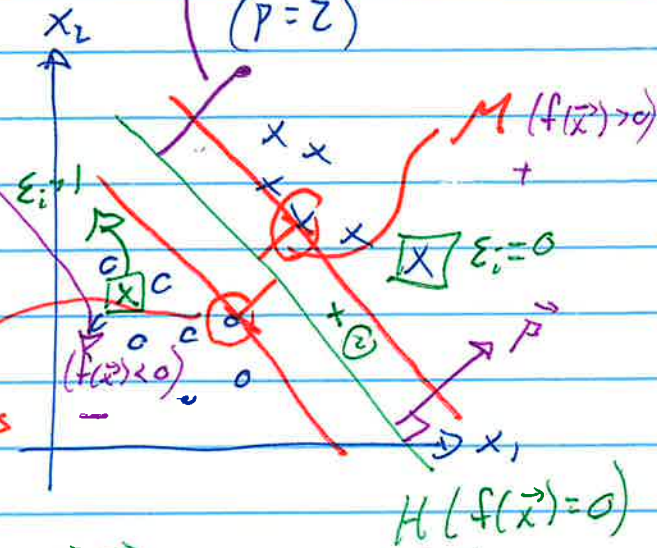
VI Max margin classifier (MMC)

Find H which

maximizes margin

$$M = \min_i \frac{|f(\vec{x}_i)|}{\|\vec{b}\|}$$

only support vectors def H



$$f(\vec{x}) = \beta_0 + \vec{\beta} \cdot \vec{x} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$$

Optimization prob:

Biggest margin

$$\max_{\beta_0, \vec{\beta}} M$$

$$M = \min_i \frac{|f(\vec{x}_i)|}{\|\vec{b}\|} \rightarrow 1$$

Normal to H

Unit normal

simplify

subject to constraints:

$$① \|\vec{\beta}\| = 1$$

$$② y_i f(\vec{x}_i) = y_i (\beta_0 + \vec{\beta} \cdot \vec{x}_i) \geq M (1 - \epsilon_i)$$

Classes on correct side

$$③ \epsilon_i \geq 0 \text{ and } \sum_i \epsilon_i \leq C \text{ (cost)}$$

Allows for train misclassif & margin violations

V2

Support vector classifier

3 cases of ϵ_i :
 ① $\epsilon_i = 0$ (right side / MMC)
 ② $0 < \epsilon_i < 1$ (margin violation, right side of H)
 ③ $\epsilon_i > 1$ (wrong side H)

C tuning param:

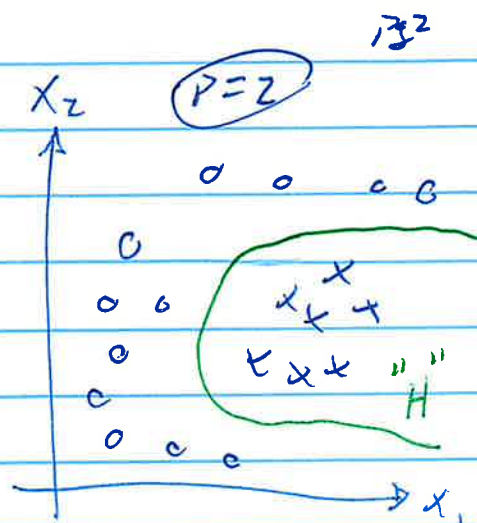
① C small (few misse)

② C big (many ")

1/3 Support vector machines

Goal: Generalize to nonlinear decision boundaries.

2 ways: ① Poly features ② Kernel trick (SVM)



Poly features: $f(\vec{x}) = \beta_0 + \vec{\beta} \cdot \vec{x} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$ (H)

$d=2$ Quadr version: $f_2(\vec{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2$
 $= \beta_0 + \vec{\beta} \cdot \vec{x}_2$, $\vec{x}_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \\ x_1 x_2 \end{bmatrix}$

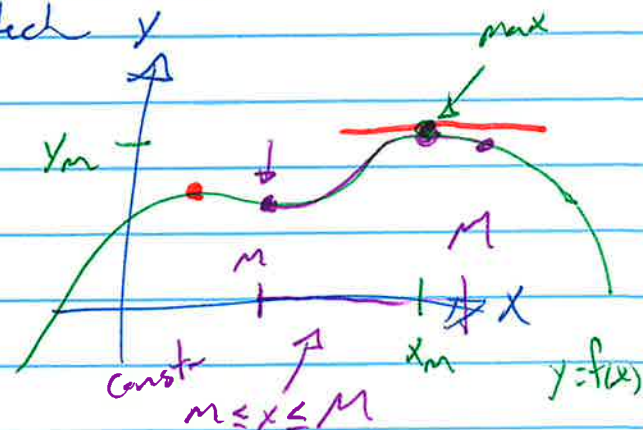
Use SVC to $f_2(\vec{x})$ & M_2 formula

$f_p(\vec{x})$ get complex for $p \gg 2$, $d \gg 2$, making this approach infeasible. (Better way!)

Kernel trick Solving optimization problem Θ uses Lagrange Multipliers from Calc 3.
 opt tech

Calc: $f'(x) = 0$
 $f''(x) < 0$ } Local max

2nd deriv test
 & check constr endpts.



Lagrange multipliers combines $\max(F(\vec{x}))$
 & constraint $G(\vec{x})$ into Lagrangian

$$L(\vec{x}, \lambda) = F(\vec{x}) + \langle \lambda, G(\vec{x}) \rangle$$

inner product
for constraint.

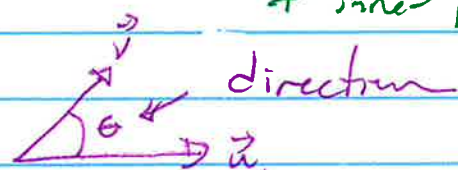
Old constraint: ② $f(\vec{x}) = \vec{\beta}_0 + \vec{\beta} \cdot \vec{x}$

Law of cosines

dot prod (simplest case
of inner product)

Recall: $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos(\theta)$

$\hookrightarrow$ compares $\vec{u} + \vec{v}$



Cosine
Similarity

$\theta = 0 \quad \vec{u} \cdot \vec{v} = 1$ (if $\|\vec{u}\| = \|\vec{v}\| = 1$)

$\theta = \pi \quad \vec{u} \cdot \vec{v} = -1$

$\theta = \pi/2 \quad \vec{u} \cdot \vec{v} = 0$

$-1 \leq \vec{u} \cdot \vec{v} \leq 1$

train data

Kernel functions: $K(\vec{x}, \vec{x}_i)$ any notion of
inner prod (vector
similar)

(SVC)

① $K_H(\vec{x}, \vec{x}_i) = \vec{x} \cdot \vec{x}_i$

Linear
decis
bound

$\hookrightarrow$ ② $\tilde{f}(\vec{x}) = \beta_0 + \sum_{i \in S} \alpha_i \vec{x} \cdot \vec{x}_i = \beta_0 + \sum_{i \in S} K(\vec{x}, \vec{x}_i)$
 only support $\rightarrow$ $i \in S$

③

Can show $\tilde{f}(\vec{x}) = f(\vec{x})$ for some $\vec{\beta}$
 involve $\vec{x}_i$ & α_i .

poly kernel



poly deg

pg 9

$$\textcircled{k2} K_p(\vec{x}, \vec{x}_i) = (1 + \vec{x} \cdot \vec{x}_i)^d, \quad d \geq 2$$

$$\tilde{f}(\vec{x}) = \beta_0 + \sum_{i \in S} \alpha_i \underbrace{K_p(x, x_i)}_{\text{Eff take of L.M. optz.}}$$

$$\left(\begin{matrix} p=2 \\ d=2 \end{matrix} \right) K_2(\vec{x}, \vec{x}_i) = \left(1 + \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix} \right)^2$$

$$\begin{aligned} &= (1 + x_1 x_{i1} + x_2 x_{i2})^2 \\ &= (1 + x_1 x_{i1} + x_2 x_{i2})(1 + x_1 x_{i1} + x_2 x_{i2}) \\ &= \boxed{1 + x_1 x_{i1}} + \boxed{x_2 x_{i2}} \\ &\quad + x_1 x_{i1} + x_1^2 x_{i1}^2 + x_1 x_{i1} x_2 x_{i2} \\ &\quad + \boxed{x_2 x_{i2}} + x_1 x_{i1} x_2 x_{i2} + x_2^2 x_{i2}^2 \end{aligned}$$

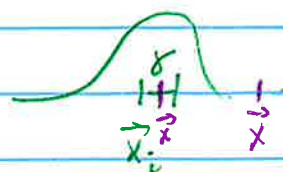
Bumps around

drawn pts
Radial Kernel

$$\begin{aligned} &= 1 + 2x_1 x_{i1} + 2x_2 x_{i2} \\ &\quad + x_1^2 x_{i1}^2 + x_2^2 x_{i2}^2 \\ &\quad + 2x_1 x_{i1} x_2 x_{i2} \end{aligned}$$

$$\textcircled{k3} K_p(\vec{x}, \vec{x}_i) = \exp\left(-\gamma \sum_{j=1}^p (x_j - x_{ij})^2\right)$$

(Gaussian) Nice 3d view $\rightarrow$ $= e^{\underbrace{-\gamma \sum_j (x_j - x_{ij})^2}_{\text{dist to train data}}}$



SVM Kernel Summary:

① $K_H(\vec{x}, \vec{x}_i) = \vec{x} \cdot \vec{x}_i$ (SVC, hyper-plane sep)

② $K_P(\vec{x}, \vec{x}_i) = (1 + \vec{x} \cdot \vec{x}_i)^d$ (poly sep, smooth curve)

③ $K_R(\vec{x}, \vec{x}_i) = e^{-\gamma \sum \cdot}$ (γ gives smooth or jagged bound)

2 add ons: ① Multiclass ($K > 2$) A vs C
 (best if K not huge) $\binom{K}{2}$ nodes $\rightarrow$ H1 OVO (one vs one) M_{AB}, M_{AC}, M_{BC}
 $\vec{x}^*$, use mjs class
 K nodes $\rightarrow$ H2 OVA (one vs all) M_{AR}, M_{BR}, M_{CR}
 $\vec{x}^*$ vs class with biggest $f(\vec{x})$
 dist from "H"

② Log reg vs SVM.
 LD No free lunch.